

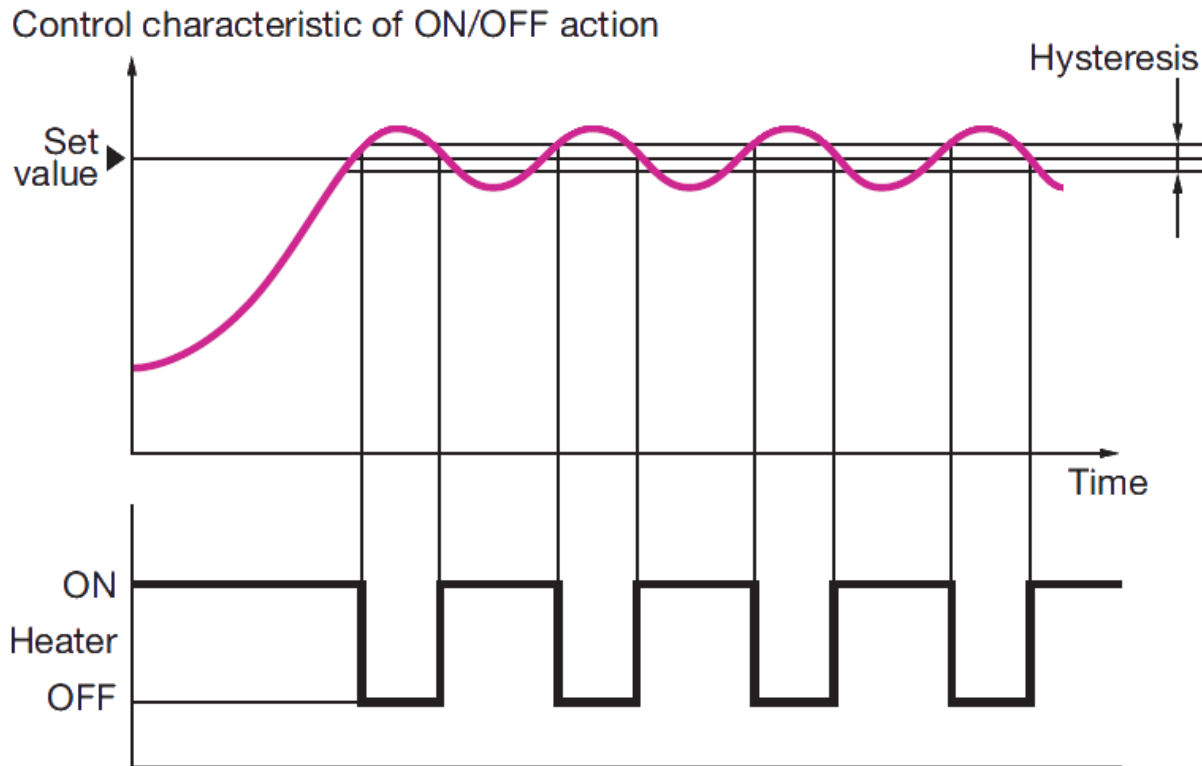
Computer Peripherals - Arduino Lab 3

1. On-Off feedback control

As a prior knowledge, a good example of these type of control can be checked [here](#).

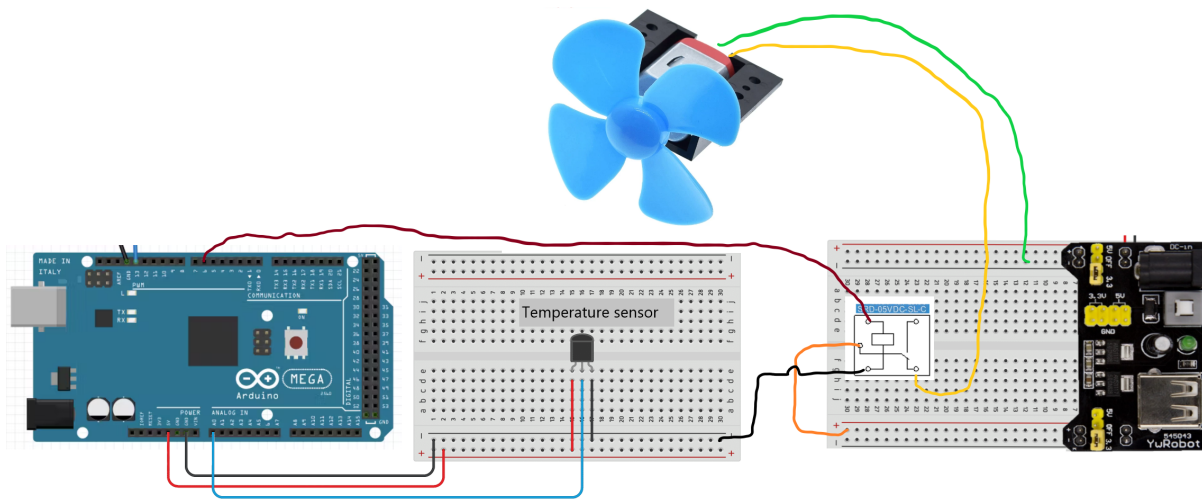
On-Off control is the simplest form of feedback control. An on-off controller simply drives the manipulated variable from fully closed to fully open depending on the position of the controlled variable relative to the setpoint. A common example of on-off control is the temperature control in a domestic heating system. When the temperature is below the thermostat setpoint the heating system is switched on and when the temperature is above the setpoint the heating switches off.

To make this on off switching, we usually use relays. To avoid fast and unnecessary switches when we are around to our setpoint we usually define a deadband or hysteresis.



2. The task

Realize the an on off control system, for temperature control. The input of the controller should be the measured temperature and the output is the on or off state of a cooling fan. A schematic can be seen on the figure below. Don't forget to define a hysteresis, good luck!



Bibliography

This laboratory guide is based on instrumentationtools.com website contents.